Industrial Experience Report

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This report includes details of my industrial experience with jobs and internships related to Computer Science.

Introduction

This report includes details my job and internship related to the field of Computer Science. My internship and job were both done at Innovative Systems Rapid City office. Innovative Systems is a leading provider of integrated solutions for the independent telecommunications industry with expertise in telecommunications, electrical engineering, embedded systems design, software engineering and marketing. My job and internship projects were done to support their IPTV solutions for customers. More specifically, I worked on developing simple games to exist on Set-Top Boxes. These games were designed for any age user to intuitively understand and interact with from their home. The following sections provide additional details and duties of my internship and job.

Internship

My internship was done at Innovative System Rapid City office during the summer of 2012. As a Software Engineer intern, my duties included:

- Develop Set-Top Box games
- Research Innovative Systems' IPTV architecture to offer insight on proof of concept for future projects
- Maintain project task management using Trello
- Maintain source code repositories using TortoiseSVN
- Follow Agile Software development methodologies

During this internship I was place on the STB Game Development team were we used Qt/C++ to develop our two projects. The first project consisted of developing and implementing a Tile Puzzle game. Tile Puzzle game is a puzzle that challenges a player to slide pieces along certain routes of a game board to establish a certain end-configuration. The puzzle was an image that was chopped into a matrix(3x3, 4x4, 5x5) of image pieces. This matrix was then randomly shuffled to create a "puzzle". A player would reach the end configuration when the image pieces were moved around to form the original image.

The second project was developing and implementing a Checkers game. Checkers was an extremely difficult game to design and implement for several reasons. First, Checkers was the first multiplayer game design to exist on a STB. It created brand new concepts to the STB Game environment such as, player's turn, local computer opponents (A.I.) with difficulty levels, and multiplayer system features. These concepts offered me unique challenges because the actual STB exists on an embedded system with extremely limited hardware and software features. Also, I had never taken a class in Artificial Intelligence which made the design of a Checkers opponent even more complicated. Also, the development environment for the STB was unstable and outdated. The unstable factor was due to the bugs in Virtual Machine environment. There would be times the Virtual Machine would just freeze and you were unable to recover data or even log back in. This made maintaining source repositories within the Virtual Machine almost impossible and overly time consuming. The STB development environment was outdated because the STB system didn't use standard Qt libraries. The provided libraries were part of a legacy system and were used to support the existing embedded hardware. Debugging was done

using line by line printf() statements and testing was done watching trace method output in a terminal window. Finally, lack of communication from senior developers created numerous design and implementation delays. These delays were due to the STB game subsystem dependencies with the main STB system. I couldn't even run my code if the main system source couldn't build and fixing the build errors and re compiling took hours. The final deliverable product had minor bugs but it was used as test game to support future senior design projects.

Job

I worked and am still employed as a Software Engineer at the Innovative Systems Rapid City Office. I had similar duties as listed in the previous section but was also assigned to the research and design of the STB multiplayer game environment proof of concept project. This project taught me many concepts that could be used in the IPTV industry. First, I learned about the overall IPTV communication architecture and how it is implemented at Innovative Systems. The architecture provided knowledge of major network hardware used at the company. Second, existing communication protocols strengthened my knowledge of how UDP and TCP protocols are implemented on real hardware with message packets that used encryption and decryption algorithms with validation. I also got familiarized with the trends of hardware and how systems are designed and deployed on them. Finally, the project taught me to develop with the end user of tomorrow in mind. Keeping the future end user in mind created many additional features that continue to support the evolving project.

Summary

Overall, I had a great experience working for my employer. The industrial experience provided unexpected growth to my skill set as a software developer that I am very thankful for. Additionally, my bosses, Brian Butterfield and Dave Springhetti, were extremely knowledgeable and professional. They created a very positive and welcoming work environment for software developers. Throughout the struggles, frustration, joy, and excitement of software engineering I had one major take away. Successful software engineers are unrealistic. They want to do something that no one has ever done before and the only way they can fail is if they give up believing in that idea. They sit there for hours, days, months, and years believing in something different. The obstacles along the way are there to signify how bad they want something because if they're willing to put in the time, who has the right to tell they're unrealistic. Was it realistic to think a computer could fit in the palm of your hand 30 years ago?